

When Is a Rule-Based Model Really Reducible?

A systems-biology guide to exact state reduction in Kappa and BNGL

Research note – less-theoretical version

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Abstract

Rule-based models are designed to avoid writing out every molecular species and reaction in systems with combinatorial complexity. BioNetGen Language (BNGL), Kappa, and network-free simulators such as NFsim let a modeler write local biochemical rules while the implied stochastic state space can be extremely large or even countably infinite [14, 19, 5]. This raises a practical question: after writing such a model, can we replace its microscopic state by a smaller *exact* state description that is still sufficient to predict a chosen experimental readout?

This note explains two limits of exact reduction using examples that resemble familiar biological mechanisms. In a processive-scanning model, a motor moves along a homogeneous chain until it reaches an endpoint. The rules are local, but exact prediction of the completion readout requires the scanner’s remaining distance, a nonlocal integer that no fixed finite collection of ordinary local Kappa/BNGL patterns can recover over all reachable restart states. In a reversible assembly-tip model, local addition and removal at a unique tip produce the classical stable M/M/1 busy-period process. From a single finite seed, the resulting completion-time law has no exact finite-state hidden continuous-time Markov chain (CTMC) realization, even though the mechanism itself has a one-dimensional integer state.

These are exactness results, not claims that finite models cannot approximate the behavior. Phase-type distributions are dense, and finite-state kinetic schemes may approximate an infinite-state mechanism arbitrarily well; however, the required model order cannot stay bounded as the approximation is driven to the exact reversible-tip law. We also give a positive boundary: if reachable molecular complexes have a uniform size bound, finitely many species-count coordinates form an exact state. Finally, we explain why no general algorithm can decide these exact-reduction properties for arbitrary finite Kappa programs, while emphasizing that useful certified subclasses and incomplete static analyses remain possible.

1 The modeling question in ordinary language

A rule-based model often contains much more information than the experiment reports. A signaling model might track receptor binding partners, phosphorylation states, scaffold occupancy, oligomer size, and subunit context, while the experiment reports only phospho-ERK. A gene-regulatory model might track promoter occupancy and chromatin states, while the assay reports RNA counts or accessibility.

Write the full stochastic molecular state as X_t and the chosen readout as

$$Y_t = h(X_t).$$

The practical reduction question is:

Can we replace X_t by a smaller state Z_t that can be simulated on its own and still reproduces the chosen readout exactly?

The word *exactly* matters. We are not asking whether a smaller model fits one dataset, whether two parameterizations are statistically indistinguishable, or whether an approximation is good enough over a particular time window. Those are important but different questions. Here the full mechanistic model is already fixed; we ask what information can be discarded without changing the stochastic behavior of a specified assay.

There are three distinct notions that are easy to conflate:

Question	Meaning	Example failure mode
Finite-state realization	Can the assay process from one chosen seed be reproduced by some finite hidden CTMC?	Completion time requires genuinely infinite state.
Finite local coordinate	Can finitely many ordinary Kappa/BNGL pattern observables supply an exact state?	A processive position is nonlocal.
Reusable reduction	Does one reduced state description remain exact for every reachable restart state?	A seed-specific mixture can collapse even though restart states cannot.

These axes are different. One of the main purposes of the examples below is to make that distinction concrete.

2 Why this issue appears naturally in rule-based biology

BNGL and Kappa were developed for biochemical systems in which local interaction rules generate combinatorially many molecular species. Instead of enumerating every receptor complex or phosphorylation configuration, a modeler specifies reactive motifs and local context [14, 5]. NFsim then simulates molecular objects and rule matches directly, avoiding prior enumeration of the full reaction network [19].

That design is extremely useful, but it creates an important distinction:

$$\boxed{\text{finite local rule set} \not\equiv \text{finite exact predictive state.}}$$

The simulator may only need local rules to advance the microscopic system, while an observer who wants to summarize the system exactly may need a global quantity such as polymer length, distance to an endpoint, or an ordered modification count.

This does *not* mean that existing Kappa reduction methods fail. Exact coarse-graining, stochastic fragments, intrinsic information carriers, and related static analyses already identify many correlations that can safely be discarded [9, 13, 10, 18]. The question here is where any finite local reduction must eventually stop.

3 Example 1: a processive enzyme that needs a nonlocal position

3.1 Biological picture

Imagine a processive enzyme moving one step at a time along a homogeneous template until it reaches a marked end. This is deliberately abstract, but it resembles the logic of a polymerase, helicase, ribosome, kinesin-like motor, or processive modification enzyme. The readout is simply whether the enzyme has finished.

The mechanism has two stages:

1. A writer extends a linear chain at rate λ_b . At rate λ_s , a Freeze event stops growth and starts scanning.
2. A unique scanner moves one monomer toward the terminal marker at rate κ . Reaching the terminal monomer produces the local **Done** readout.

Every reaction rule only needs a small local neighborhood. A Kappa rule can specify the scanner, its adjacent monomer, and the relevant bond. The analogous BNGL construction uses molecule components, internal states, and explicit bonds.

If the scanner has r steps left, then completion requires r independent rate- κ events. The completion time is therefore Erlang(r, κ), with survival probability

$$S_r(t) = e^{-\kappa t} \sum_{j=0}^{r-1} \frac{(\kappa t)^j}{j!}.$$

Different values of r give different future readout distributions.

3.2 Why ordinary local patterns cannot recover position

Suppose we choose any finite collection of ordinary Kappa or BNGL patterns. Every pattern has finite size, so collectively they can inspect only a bounded neighborhood around the scanner and bounded local motifs elsewhere in the chain. Now make the chain long and place the scanner at two different positions that are both far from either endpoint.

Locally, the two states look identical to every chosen pattern. The chain is homogeneous around the scanner, and the boundary markers lie outside the patterns' reach. Therefore the entire finite pattern-observable vector can be identical in the two states. But the scanner can still be, say, 20 steps from completion in one state and 200 steps away in the other. Their future Done-time laws are different.

The formal statement is:

Theorem 3.1 (Local-pattern insufficiency, informal statement). *For every finite family of finite ordinary site-graph patterns, there are two reachable scanner states with identical values for all those patterns but different future laws of the Done assay.*

A proof is given in [section B](#). The key fact is purely spatial: finite patterns have bounded reach, while the relevant coordinate is an unbounded distance.

3.3 The practical lesson

The system *does* have a simple exact state:

$$Z_t = \text{distance remaining.}$$

That is just one integer. The problem is not that the exact state is high-dimensional; the problem is that this integer is *nonlocal* and cannot be reconstructed from any fixed finite vocabulary of ordinary local patterns over all reachable states.

This gives a clean interpretation of why explicit counters are useful in rule languages. Kappa's counter extension supplies an integer-valued coordinate directly, which is exactly the type of state ordinary bounded-neighborhood patterns cannot express [6].

Important subtlety: the original seed can still collapse. The scanner’s particular construction seed is much easier than the collection of all restart states. Growth stops at a geometrically distributed chain length, followed by a one-way Erlang scan. That mixture simplifies to a two-phase completion-time law, so the binary assay from the original seed has an exact three-state hidden CTMC realization (two transient states plus Done). Thus:

hard to reduce for every restart state $\nRightarrow$ hard to realize from one seed.

The exact calculation is in [section B](#).

4 Example 2: a reversible assembly tip with no finite-state exact realization

4.1 Biological picture

Now consider a rooted polymer or oligomer with one distinguished reactive tip. Two local events are possible:

$$r \rightarrow r + 1 \quad \text{at rate } \lambda, \quad r \rightarrow r - 1 \quad \text{at rate } \mu,$$

where r is the number of active units and $\mu > \lambda$. When the last active unit disappears, the assay switches permanently to Done.

A unique tip marker is important. It makes the total Push and Pop hazards independent of polymer length: there is exactly one reactive tip, hence exactly one Push match and one Pop match at any positive length. Without a distinguished tip, ordinary mass-action counting would generally make a removal hazard scale with the number of matching units.

The length process is the same birth–death chain that describes the classical stable M/M/1 queue busy period, started with one customer [20]. The queueing result is not new; what matters here is that ordinary local biochemical site-graph rules can realize it.

4.2 Why no finite hidden CTMC can be exactly equivalent

Let T be the time until the active length first reaches zero. Its Laplace transform is

$$\rho(s) = \mathbb{E}[e^{-sT}] = \frac{s + \lambda + \mu - \sqrt{(s + \lambda + \mu)^2 - 4\lambda\mu}}{2\lambda}.$$

The square root makes this transform non-rational.

By contrast, any finite-dimensional time-homogeneous linear kinetic system produces observables of the form

$$u^T e^{Gt} b,$$

whose Laplace transforms are rational functions because they contain $(sI - G)^{-1}$. Finite hidden CTMCs are a special case of this matrix-exponential form [4]. Therefore no finite hidden CTMC can reproduce this completion assay *exactly* from the finite seed.

Theorem 4.1 (Single-seed realization limit, informal statement). *The reversible-tip Done process from one active monomer has no exact finite-dimensional time-homogeneous linear realization, and hence no exact finite-state hidden CTMC realization.*

The proof is in [section C](#).

4.3 What this does and does not mean for fitting data

This theorem is about exact representation, not empirical distinguishability. Phase-type distributions—the absorption-time distributions of finite CTMCs—are dense among distributions on $[0, \infty)$ [3, 4]. Thus a sufficiently large finite-state model can approximate the busy-period law arbitrarily well.

There is still a structural consequence: the model order cannot remain bounded as the approximation is driven to the exact law. Any convergent sequence of finite phase-type approximations must use an unbounded number of phases. The exact completion density also has the characteristic long-time form

$$f_T(t) \sim C t^{-3/2} e^{-(\sqrt{\mu} - \sqrt{\lambda})^2 t},$$

whereas finite matrix-exponential models contain integer-power polynomial factors multiplying exponentials. This is useful as intuition for why increasing model order may keep being demanded, but it is *not* a claim that finite experimental data can falsify every finite-state model.

5 A positive result: when finite exact coordinates are guaranteed

The examples above depend on unbounded connected structure. There is a broad and simple positive case.

Theorem 5.1 (Bounded-complex certificate, informal statement). *If there is a fixed bound B on the number of agents in every connected molecular complex that can ever be reached, then finitely many fully specified connected-species counts form an exact finite-dimensional coordinate system.*

The reason is straightforward. With finitely many molecule types, sites, and internal states, only finitely many connected species graphs of size at most B exist up to isomorphism. The full mixture is therefore determined by a finite vector of their copy numbers. Copy numbers can still be arbitrarily large, so the resulting state space may be infinite, but the *coordinate dimension* is finite.

This distinction matters:

Property	Meaning
Finite rule syntax	The model file contains finitely many molecule types and rules.
Finite coordinate dimension	A fixed finite vector of numbers describes the exact state.
Finite state cardinality	Only finitely many distinct state values are reachable.
Finite local expressibility	The coordinate can be computed from finitely many ordinary local patterns.

The reversible tip has finite coordinate dimension (one integer) but infinite state cardinality and no finite-state assay realization. The scanner also has a one-integer exact coordinate, but that coordinate is nonlocal.

6 How this relates to existing Kappa/BNGL reduction methods

A large literature already studies exact and approximate reduction of biochemical models. In Kappa, internal coarse-graining, abstract differential semantics, intrinsic information carriers, stochastic fragments, and lumpability analyses exploit the fact that a rule usually depends on only a small

part of each molecule [9, 7, 13, 10]. Petrov’s survey summarizes how Markov-chain aggregation has been applied to rule-based modeling [18]. In deterministic kinetic models, CLUE similarly finds exact constrained lumpings that preserve user-selected observables [17].

The present results should be read as *limits around those positive methods*, not replacements for them. They say that there are finite local rule programs for which:

- no finite set of ordinary local pattern observables can be a reusable exact predictive state; or
- even a completely unrestricted finite hidden CTMC cannot exactly reproduce a selected assay from one seed.

That is a stronger obstruction than “the full reaction network is too large to enumerate,” which is exactly the problem network-free methods such as NFsim were designed to avoid [19]. Network-free simulation can simulate an infinite or combinatorial microscopic process without implying that a finite exact reduced state exists.

7 Why there cannot be a perfect automatic checker

A natural software goal would be: feed in a Kappa model and an assay, and have static analysis answer whether a finite exact reduction exists. Existing Kappa static analysis makes this goal plausible: the platform performs reachability-style analyses and reasons about structural properties of rule systems [5]. But a sound-and-complete checker cannot exist for arbitrary finite Kappa programs.

The technical reason is that an ordinary free-site match acts as an exact zero test. A counter can be encoded as a chain attached to a unique root: adding one unit is a local rule, removing one unit is a local rule, and testing whether the counter equals zero is simply asking whether the root’s counter site is free. Two such counters plus a finite controller can simulate a deterministic two-counter machine. Halting for the corresponding machine model is undecidable [16, 8].

The formal reductions use the two examples above as assay modules. They establish two undecidable decision problems:

1. whether the assay from a designated finite seed has any exact finite hidden-CTMC realization; and
2. whether some finite family of local patterns gives a reusable exact Markov state.

The proofs are Rice-style in architecture: a universal computation is routed into behaviors on opposite sides of the property. The nontrivial part is enforcing stronger promises so the result is not merely “halting gives a finite system, nonhalting gives an infinite one.” In the constructions, both YES and NO branches have assay-relevant infinite reachable behavior and countably infinite observation-compatible strong quotients; the local-pattern reduction can additionally be kept under a uniform exit-rate bound. Details are summarized in [section E](#).

There is an instructive comparison with pushdown systems. For an ordinary pushdown configuration, Jančar proved that deciding whether it is bisimilar to some finite-state process is decidable; Göller and Parys later showed that a bisimulation-finite pushdown system has an equivalent finite system of elementarily bounded size, yielding elementary decidability [15, 12]. Kappa’s free-site zero test crosses from one-stack-like memory to two-counter computational power. This comparison does not make our stochastic realization questions strict generalizations of pushdown semantic finiteness, but it shows that undecidability is not a generic consequence of having an unbounded chain.

The correct practical conclusion is not “static analysis is hopeless.” It is narrower:

No analysis can be both sound and complete for these exact-reduction questions over arbitrary finite Kappa programs. Useful sufficient conditions, certified subclasses, and incomplete diagnostics are still possible—and are exactly what one should build.

The bounded-complex theorem is one such sufficient condition.

8 What this means for signaling models

The examples are intentionally stripped down, but the hidden coordinates they expose have direct biochemical analogues.

Processive modification. If an enzyme walks through an ordered sequence of sites, an assay may depend on how many steps remain rather than on any bounded local motif. Examples include processive phosphorylation/dephosphorylation, sequential assembly, or template traversal. A rule-based model can represent every step locally while exact prediction still requires a position counter.

Receptor clusters and scaffolds. If signaling depends on cluster size or the size of an active connected component, the relevant state may be an unbounded integer even when every bond-forming and bond-breaking rule is local. The model may be easy to run in NFsim but impossible to replace exactly by a fixed finite-state kinetic scheme.

History hidden in structure. Two molecular mixtures can have identical values for all observables a modeler happened to write down and yet have different future distributions because the observables omitted an ordered or nonlocal structural coordinate. This is a structural form of non-Markovianity induced by projection: the full rule-based process is Markov, but the chosen observable vector need not be.

For a modeler, this suggests that “is my observable vector Markov?” is a more useful question than “did I include the important proteins?” A biologically sensible observable list can still omit the state variable required for exact closure.

9 What this means for single-cell RNA/ATAC modeling

The theorems in this note are stated for deterministic local assays of a rule-based molecular state. Single-cell data introduce an additional observation layer—for example, transcription and chromatin states may generate stochastic UMI or fragment counts. Nevertheless, the same modeling distinction remains useful.

A conceptual pipeline is

mechanistic state X_t $\longrightarrow$ latent assay-relevant state $\longrightarrow$ RNA/ATAC measurement.

A finite-dimensional measurement does not imply that the latent mechanistic state needed to predict its distribution is finite-state or locally expressible. For example, an enhancer or promoter readout could depend indirectly on receptor-cluster size, an ordered modification count, or a processive chromatin mechanism.

This matters especially when a mechanistic simulator is followed by an observation operator and statistical inference. If a reduced mechanistic state is assumed to be finite or Markov solely

because the observed assay vector is finite, that assumption needs justification. The alternatives are legitimate but should be explicit:

- add the missing nonlocal state (for example, a counter or cluster size);
- retain an infinite-state/network-free mechanistic representation; or
- make an approximation and validate its error for the intended assay and time horizon.

Nothing here says that single-cell data can identify the full microscopic mechanism. Model identifiability is a separate problem: multiple mechanistic models may generate effectively indistinguishable data. The question addressed here is conditional on a chosen mechanistic model and asks whether *that model* has an exact smaller state representation for a specified readout.

10 A practical workflow for rule-based modelers

The theory suggests a concrete workflow when trying to reduce a Kappa/BNGL model.

1. **Fix the readout first.** Exact reduction is observation-relative. “Preserve the model” is too vague; specify the quantities or assay process that must remain exact.
2. **Separate seed-specific from reusable goals.** If the model will only be used from one experimental initial condition, a seed-specific hidden realization may be much smaller than any reduction valid after arbitrary restarts or perturbations.
3. **Check for obvious unbounded structural coordinates.** Polymer length, cluster size, distance along a template, number of completed processive steps, and rooted stack depth are warning signs.
4. **Test closure, not just biological relevance.** Two states with the same proposed reduced coordinate should have the same aggregate rates into every reduced successor state. If not, the coordinate remembers too little. This is the rule-level form of classical lumpability [10, 11].
5. **Use counters when the biology is genuinely count-like.** A single integer may be a much better exact state than dozens of local motifs. Kappa counters were developed precisely to represent quantities that are awkward to encode by repeated site-graph structure [6].
6. **Distinguish exactness from approximation.** Failure of finite-state exactness does not make finite-state inference useless. It means model order is an approximation choice, not a hidden exact mechanistic truth.
7. **Exploit positive structural certificates.** If static analysis or biological constraints guarantee bounded connected-complex size, a finite exact species-coordinate system exists.

11 Take-home message

The central point can be stated without formal-language terminology:

Local biochemical rules can generate dynamics whose exact predictive state is global, unbounded, or impossible to represent by any finite-state kinetic model.

The processive scanner shows the first failure: local patterns do not necessarily tell you where you are in a long homogeneous structure. The reversible tip shows the second: even from one finite seed, a simple local mechanism can produce an assay law outside every finite-state hidden CTMC class. Neither result says that practical approximations are impossible. Rather, they tell us what an approximation is approximating and why adding more local observables or more hidden states may never produce a fixed exact finite model.

For systems biology, that is the useful boundary: before interpreting a reduced model as the exact mechanistic state, ask whether the underlying rule system contains an unbounded structural coordinate that the proposed state has silently thrown away.

A Technical setup: rule-based CTMCs and exact reduction

The main text deliberately suppresses formal machinery. This appendix states the assumptions used by the examples.

A finite Kappa-style signature has finitely many agent types, sites, internal states, and finite rewrite rules. Starting from a finite seed, each reachable mixture is a finite site graph, although the number of reachable mixtures may be countably infinite. Under the usual stochastic semantics, a rule r has a finite set of matches m in a mixture x ; applying a match gives a new mixture $T_r(x, m)$ with stochastic hazard $a_r(x, m)$. The generator is

$$(\mathcal{L}f)(x) = \sum_r \sum_{m \in \text{Match}_r(x)} a_r(x, m) [f(T_r(x, m)) - f(x)].$$

This formulation absorbs software-specific symmetry and rate normalization into the exact match hazard. Kappa’s stochastic graph-rewrite semantics and BNGL/NFsim’s rule-based stochastic execution fit this local-match picture for the constructs used in the examples [5, 14, 19].

For a proposed coordinate $\Phi(x)$, define the total rate from x to reduced value z' by summing the hazards of all rule matches whose products have $\Phi = z'$. Classical strong lumpability says that $\Phi(X_t)$ is a time-homogeneous Markov process for every initial state exactly when these aggregate rates depend on x only through $\Phi(x)$ [10, 11]. This is the closure test invoked in the practical workflow.

B Proof details for the processive scanner

B.1 Local-pattern theorem

Let F be a finite family of finite site-graph patterns. For each connected pattern P , any embedding occupies a finite graph diameter. On a sufficiently long homogeneous chain, two scanner positions far enough from both boundaries have isomorphic neighborhoods large enough to contain every scanner-touching embedding of every $P \in F$. Embeddings that do not touch the scanner are unchanged in total number by translating the unique scanner defect through the homogeneous interior. Any fixed ambient mixture contributes equally to both states. Therefore all connected-pattern counts in F agree.

Disconnected finite-pattern counts can be written as finite polynomial combinations of connected-pattern counts through the standard overlap/gluing identities for graph observables used in rule algebra [1, 2]. Hence every pattern count in F agrees as well. Choosing two deep positions with distinct remaining distances $r \neq r'$ gives identical local statistics but different Erlang survival functions $S_r \neq S_{r'}$. Thus no finite local pattern family determines the future Done law over all reachable restart states.

B.2 Why the designated seed has only two transient phases

Let $F_n(s)$ be the Laplace transform of the Done time while the system is still in build mode with chain length n . Define

$$u_b = \frac{\lambda_b}{\lambda_b + \lambda_s + s}, \quad v = \frac{\lambda_s}{\lambda_b + \lambda_s + s}, \quad w = \frac{\kappa}{\kappa + s}.$$

First-event conditioning gives

$$F_n = u_b F_{n+1} + v w^n.$$

The bounded solution is $F_n = v w^n / (1 - u_b w)$, so from the one-monomer seed

$$\mathbb{E}[e^{-sT}] = \frac{\lambda_s \kappa}{s^2 + (\lambda_b + \lambda_s + \kappa)s + \lambda_s \kappa}.$$

The denominator has two positive roots (or a repeated positive root in a limiting degenerate case), so this is the transform of a two-phase phase-type distribution. Adding the absorbing Done state gives a three-state exact hidden CTMC for the binary assay from that seed.

C Proof details for the reversible tip

With one unique active tip, there is exactly one Push match and one Pop match at each positive active length. Therefore active length is a birth-death chain

$$r \rightarrow r+1 \text{ at } \lambda, \quad r \rightarrow r-1 \text{ at } \mu,$$

with 0 absorbing and $\mu > \lambda$. If $f_r(s) = \mathbb{E}_r[e^{-sT}]$, first-step analysis gives

$$(s + \lambda + \mu)f_r = \lambda f_{r+1} + \mu f_{r-1}, \quad f_0 = 1.$$

Translation invariance and the strong Markov property give $f_r = \rho^r$, where

$$\lambda \rho^2 - (s + \lambda + \mu)\rho + \mu = 0.$$

The bounded root with $\rho(0) = 1$ is

$$\rho(s) = \frac{s + \lambda + \mu - \sqrt{(s + \lambda + \mu)^2 - 4\lambda\mu}}{2\lambda}.$$

The discriminant has two distinct simple roots at $-(\sqrt{\mu} - \sqrt{\lambda})^2$ and $-(\sqrt{\mu} + \sqrt{\lambda})^2$, so its square root is not rational. Hence ρ is not rational.

For the survival function $S(t) = \mathbb{P}(T > t)$,

$$\hat{S}(s) = \frac{1 - \rho(s)}{s}$$

is also non-rational. Any finite-dimensional time-homogeneous linear realization would have $S(t) = u^\top e^{Gt} b$, yielding the rational transform $u^\top (sI - G)^{-1} b$, a contradiction. This proves the single-seed nonrealizability statement. The busy-period formula is classical [20]; the matrix-exponential implication is standard [4].

The density is

$$f_T(t) = \sqrt{\frac{\mu}{\lambda}} \frac{1}{t} e^{-(\lambda+\mu)t} I_1(2t\sqrt{\lambda\mu}),$$

and $I_1(z) \sim e^z / \sqrt{2\pi z}$ gives the $t^{-3/2}$ prefactor. Phase-type denseness is classical [3]. The bounded-order obstruction follows because phase-type Laplace transforms of uniformly bounded order are rational functions of uniformly bounded degree; a coefficient-normalized convergent subsequence would force the non-rational transform ρ itself to be rational.

D Bounded complexes and finite exact coordinates

Assume every reachable connected complex contains at most B agents. Because the agent signature is finite, there are only finitely many fully specified connected species graphs $C_1, \dots, C_M$ of size at most B up to isomorphism. A mixture is uniquely specified by

$$\Phi(x) = (n_{C_1}(x), \dots, n_{C_M}(x)) \in \mathbb{N}^M.$$

Thus Φ is injective and is automatically an exact Markov coordinate. The copy-number state space can remain countably infinite, but the coordinate dimension is finite.

E Undecidability: what the reduction actually uses

A deterministic two-counter machine has a finite control state and two nonnegative integer counters. The relevant instruction forms increment a counter or perform a zero-test/decrement branch. Halting is undecidable for the CM2 instruction model used in the reduction [8]; two-counter universality goes back to Minsky [16].

In ordinary Kappa, represent one counter as a chain attached to a unique root. Increment locally inserts one node at the root end. A nonzero decrement locally removes that first node. The zero-test branch matches the same root site only when it is free. Because the free and bound conditions are mutually exclusive, exactly one branch is enabled. A finite controller agent stores the machine instruction. This gives an exact deterministic simulation of the machine’s unique run at the level of enabled rule structure; the reduction does not rely on approximate stochastic computation.

For the seed-level decision problem, a probe can be coupled to the simulated step count so that nonhalting yields a non-rational one-time survival transform (with infinitely many poles), whereas halting switches into the processive scanner, whose designated seed has a finite phase-type realization. The construction is arranged so that both branches still have assay-relevant infinite behavior and countably infinite observation-compatible strong quotients.

For the reusable local-pattern decision problem, the halting branch switches into the reversible-tip module, which has a simple one-counter exact coordinate; the nonhalting branch exposes arbitrarily long frozen scanner states, so the local-pattern theorem applies. Again, both branches are infinite in the assay-relevant sense. These strengthened promises are why the result is more informative than the trivial observation that a Turing-complete language has undecidable semantic properties.

For context, semantic finiteness of ordinary pushdown processes behaves differently: Jančar proved decidability of whether a pushdown configuration is bisimilar to some unspecified finite-state process [15], and Göller and Parys proved an elementary bound on the size of an equivalent finite system when one exists, yielding elementary decidability [12]. This comparison motivates the decision question but is not a strict generalization claim; the equivalence notions and probabilistic realization problem differ.

F Kappa/BNGL construct correspondence

The proofs are written in Kappa-style site-graph language because it has a clean formal graph-rewriting semantics. The examples use only standard constructs with direct BNGL/NFsim analogues:

Ingredient	Kappa	BNGL/NFsim
Typed molecular object	agent signature	molecule type
Internal marker	site internal state	component internal state
Polymer adjacency	explicit link	explicit component bond
Free-site test	site required unbound	unbound component in reactant pattern
Local synthesis	agent appears on RHS	molecule appears in product pattern
Local assay	pattern observable	Molecules observable
Network-free simulation	KaSim	NFsim

BNGL also provides **Species** observables and functional expressions. The scanner locality result is not evaded by applying a fixed deterministic function to finitely many local observables: if the inputs are identical in two deep-interior states, the function is identical too. On the single scanner complex, a finite set of finite-pattern **Species** tests also cannot recover unbounded position. An explicit nonlocal counter is outside the theorem’s restricted local-pattern vocabulary.

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